

ROI Analyses Left Amygdala - Model 1 (Encode & Recall)

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Amygdala")
ROIdata <- read.csv("BOLD_LAmygdala_model1.csv")

# Assign factors
ROIdata$S_ID <- factor(ROIdata$S_ID)
ROIdata$ID <- factor(ROIdata$ID)
ROIdata$StimType <- factor(ROIdata$StimType)
ROIdata$TaskStage <- factor(ROIdata$TaskStage)
```

Sham Condition - left amygdala

```
# One sample t-test - Sham Encode, Left hemisphere - Table S10

# BOLD = median % BOLD signal

Sham_enc <- subset(ROIdata, StimType=="Sham" & TaskStage=="Encode" )
type = "greater"
Sham_ost <- ddply(Sham_enc, .(TaskStage), t.test.plyr, "BOLD")
Sham_ost <- ost_FDR(Sham_ost)
pander(Sham_ost)
```

Table 1: Table continues below

TaskStage	statistic	p.value	estimate
Encode	5.25015333402699	2.2804493404787e-05	0.240053

Table 2: Table continues below

conf.int1	conf.int2	nobs	dof	method	alternative
0.160991781708731	Inf	20	19	One Sample t-test	greater

null.value	sig	FDR
0	**	2.28e-05

Stim Conditions - left amygdala - Figure S5

Does stimulation impact BOLD activity during encode vs recall stages of the task in the left amygdala?

```
BA <- lmer(BOLD ~ StimType*TaskStage + (1|S_ID), data = ROIdata, REML = TRUE)
Anova(BA, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##
##           F Df Df.res    Pr(>F)
## StimType    1.6457  2    95    0.1983
## TaskStage   39.7373  1    95 9.081e-09 ***
## StimType:TaskStage 1.7717  2    95    0.1756
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Plot
p.BA.amyg_L <- ggplot(ROIdata, aes(x = StimType, y = BOLD, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.4) +
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.6, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Sham","TI 1:1","TI 1:3"),
    labels=c("Sham","TI 1:1","TI 1:3")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("") + xlab("Left Amygdala")
```

```
ggsave(p.BA.amyg_L,file="Figure_S5.tiff", width = 9, height = 9, dpi = 300, units = "cm", device = "tiff")
p.BA.amyg_L
```

